


# MSc in Computer Science

Interim Report

Group 1 – Find your Dublin

Vikneshwaran Venkatesh – A00047404

Santhosh Sekar Asha – A00047302

Sowmiya Madhan Kumar – A00047022

Catherine Sherina David Chelladurai – A00046743

Shreyash Chandradip Salunkhe – A00047285

1. User Scenario ..... 3
1.1 Who is your target user? ..... 3
1.2 Why are they important? ..... 3
1.3 What problem are you solving? ..... 3
2. Technical Problem: The Setting ..... 5
2.1 Why does the system Exist? ..... 5
2.2 What is the core technical problem? ..... 6
2.2.1 Data integration ..... 7
2.2.2 Multi-criteria decisions ..... 7
2.2.3 Rental evidence ..... 8
2.2.4 Geographic accuracy ..... 8
2.2.5 Explainability ..... 8
2.3 Existing Systems and their Limitations ..... 9
2.3.1 Daft.ie ..... 9
2.3.2 Google maps ..... 10
2.3.3 Numbeo ..... 11
2.4 Requirement specification ..... 13
3. Technical Solutions: ..... 16
3.1 What does the system do? ..... 16
3.2 System Architecture ..... 16
3.3 Front end ..... 17
3.4 Back end ..... 18
3.5 Data collection and Preparation ..... 18
3.6 Prototype ..... 19
4. Evaluation ..... 23
4.1 What does success look like for your system? ..... 23
4.2 How will you evaluate the system that you built? ..... 23
5. Conclusion: The Plan ..... 26
References ..... 28

# 1. User Scenario

## 1.1 Who is your target user?

The Primary target user of our project will be those international students who are either planning to relocate to Dublin for study purpose or have just relocated to Dublin. The international students will have to choose suitable locations depending upon their budget, choice of universities, transportation and accessibility of other amenities.

Students who are unfamiliar with Dublin may find it difficult to compare districts and identify neighborhoods that suit their budget, university location, commuting requirements and lifestyle preferences. Guidance on student accommodation can help students understand their options when moving to Dublin (Technological University Dublin, n.d.). Secondary target users include prospective students, current students who are planning to relocate within Dublin, and other individuals who wish to compare residential areas based on their personal requirements.

It depends upon their preference as well as on various other factors such as accommodation style budget and transportation. The secondary target users of this system include the future and present students who want to move within Dublin and others who require help in comparing the neighborhoods of Dublin.

## 1.2 Why are they important?

International students are a particularly valuable target audience because securing proper housing is a major issue for them when considering education in another city. The rental fees should be combined with time spent travelling, proximity to public transportation, grocery stores, cafes, medical centers, parks, and many other services.

Those international students who are unfamiliar with Dublin do not know what areas provide the right balance between the cost of living and accessibility. It may take a lot of time to find out that information and analyze it. Rental statistics published by the Central Statistics Office and the Residential Tenancies Board provide useful context for understanding rental costs in Ireland (CSO, 2023; RTB, n.d.).

Understanding students needs is essential because their budgets and priorities differ. Our team collected student survey responses to identify the factors that influence accommodation decisions. These findings help inform the recommendation system's design and ensure that its features reflect user's preferences.

## 1.3 What problem are you solving?

Our project aims to address this problem by developing a Dublin District Recommendation System that brings relevant district information together on one platform. Users can enter their university location, accommodation budget, preferred accommodation type and the importance of different amenities.

Based on these preferences, the system ranks suitable Dublin districts and presents the top recommendations with match percentages and explanations. This helps users understand why particular districts may suit their requirements and compare the available options more effectively.

For example, an international student studying at TU Dublin Grangegorman with a limited budget may prioritise affordable accommodation, nearby supermarkets and gyms. By entering these preferences, the student can compare the recommended districts and identify which areas best match their needs.

Rental data published by the Residential Tenancies Board can provide a source for rental information used in district comparisons, where the data matches the property types and geographical areas being analysed (RTB, n.d.).

Student survey responses help identify the factors that influence accommodation decisions and inform the design of the recommendation system to better reflect users' needs and preferences.

# 2. Technical Problem: The Setting

## 2.1 Why does the system Exist?

Ireland welcomes lots of international students every year. According to the Higher Education Authority (HEA), there were 278,880 student enrolments in 2024/25, and 44,535 of them were non-Irish-domiciled, a rise of 10.2% on the previous year. Higher Education Authority, 2025 reveals that India was the most common country of domicile at 20.6%. Many of these students require practical assistance on arrival.

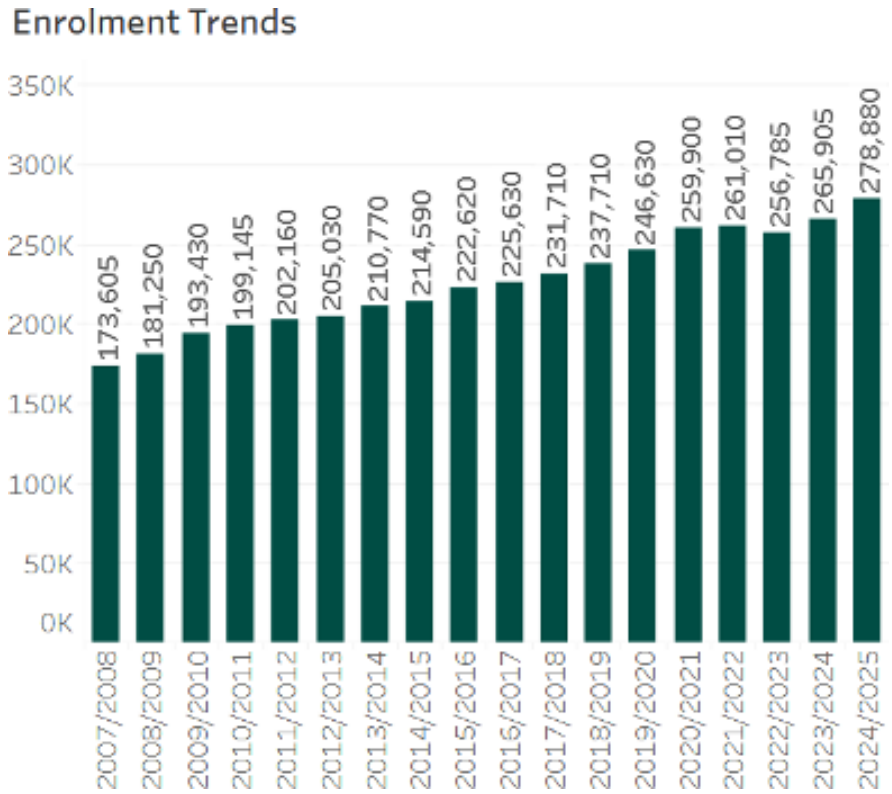



*Figure 1: Visualisation showing Enrolments trends through various timeline (HEA, 2025)*




*Figure 2: Heatmap describing the top three countries of non-Irish enrolments (HEA, 2025)*

There is much more to moving abroad than just selecting a university. Students must also make a choice of where they will live, sometimes before they are aware of the neighbourhoods, transport, and rent prices of the city. In September 2025, RTÉ News reported on the case of one international student who was looking for accommodation in Dublin, where high rents, potential scams, long commutes and queries that were never answered are all a problem (Connor, 2025). The wrong area can result in long commuting times, additional travel expenses, or limited access to services and facilities that are essential every day.

There is information available online, but it's found on various platforms: property websites for listings, maps for routes, and others for local amenities. It's difficult to mix, particularly for someone who hasn't had the privilege of being in Dublin. Find Your Dublin takes a different approach by matching postal areas based on your budget, university, type of living and lifestyle preferences. It does not give recommendations for properties. It emphasizes areas and why a district may be a good fit for a student.

## 2.2 What is the core technical problem?

The primary technical challenge is to convert multiple datasets relating to Dublin districts into a set of recommendations that are both personalised and explainable. The project covers 21 postal districts (D1–D18, D20, D22 and D24). It combines the amenity data (supermarkets, restaurants, cafés, gyms, parks, pharmacies, hospitals and shopping malls) with the rental and commute data for universities.

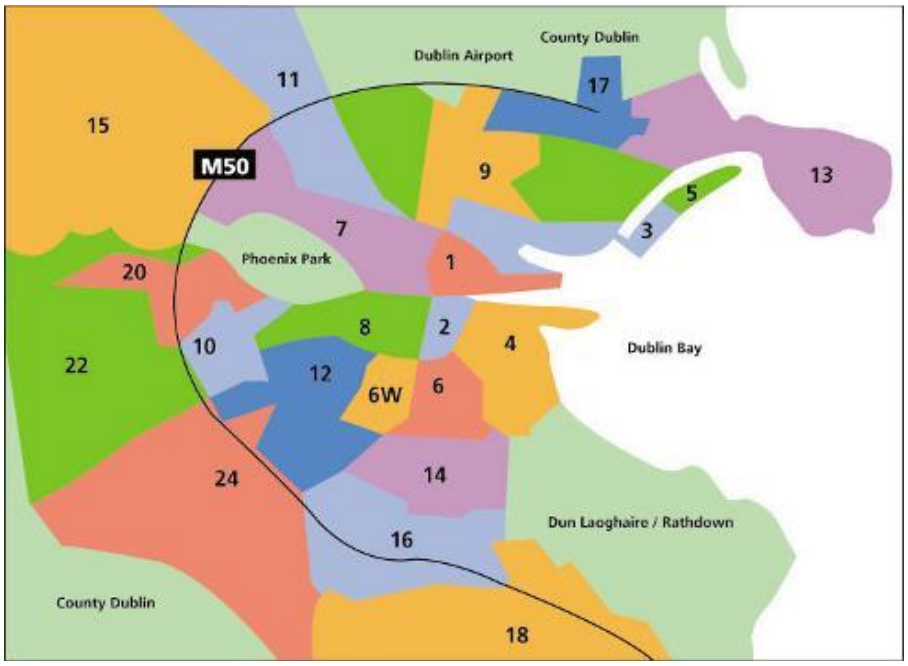



*Figure 3: Map of Dublin postal districts used in the project (dublinmap360.com)*

### 2.2.1 Data integration

The amenity data is from OpenStreetMap, the rent data from RTB, Rent.ie and Daft.ie listings and the commute data is from a routing tool manually retrieved; hence they will vary in coverage, date of collection and how they are collected. Next, a missing value should not be considered zero as that may represent that the data is not available, and not that the facility does not exist. In the project, missing values are omitted from the calculation of the normalization and are not replaced with a fabricated score but simply show missing. The amenity counts are only those that have been mapped and thus depend on the completeness of the data.

### 2.2.2 Multi-criteria decisions

Students have various needs. A fixed ranking wouldn't work because one might be more concerned with the rent, another with commuting time, and another with parks/gyms. The project uses a weighted score model. Each criterion is scaled on a 0-100 scale; weights are determined by the priorities assigned by the student, and the scores are summed to create a district score. The amenities are divided into three groups, and each group score is the mean of its components, thus students are not required to rate 10 individual amenities. A high amenity score doesn't necessarily indicate that the district is the best choice; it simply indicates that the district has a high count of amenities.

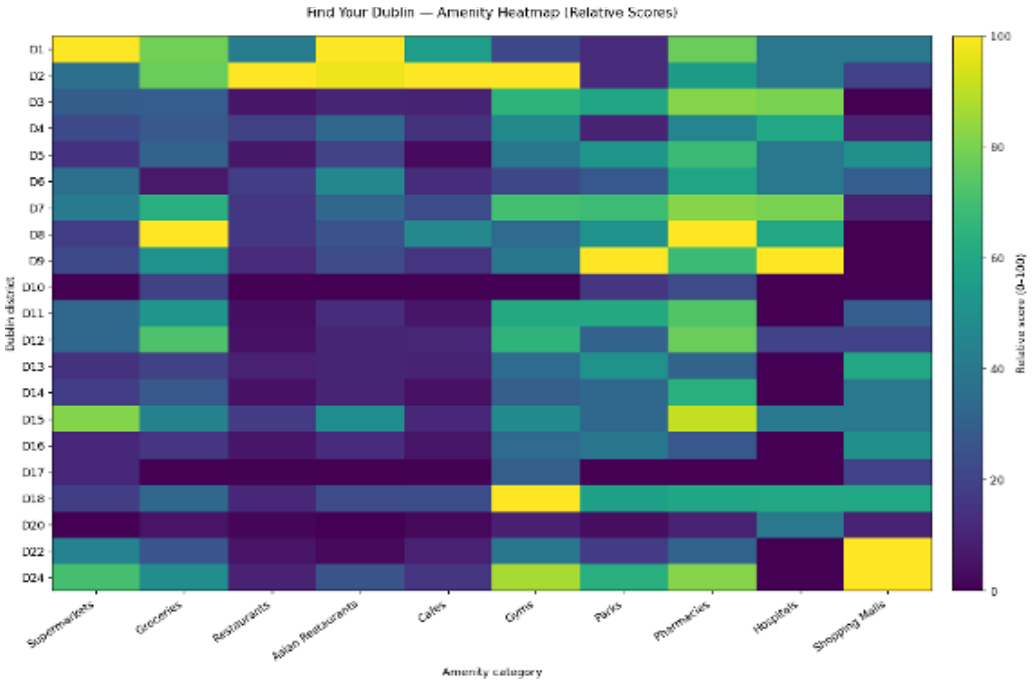



*Figure 4: Relative amenity scores across the 21 Dublin districts*

### 2.2.3 Rental evidence

Direct comparisons are not possible as rental data is not available for all accommodation types across all districts. Rental figures can change according to the type of rental, number of records, and date of data collection. Limited records may not be a good indicator of the broader rental market. If there is no rental information, it must not be interpreted as a zero, but as an indication that there is no evidence of a rental. Thus, the data should be used as a guide to determine if a district is a good match for a student's budget.

### 2.2.4 Geographic accuracy

The distance is not a good indicator of how easy it will be to travel to get somewhere. The project applies walking distance and time to all universities using route-based data from OpenTripPlanner based on OpenStreetMap data. These are indicative, not guaranteed journey times and may differ within a postal district.

### 2.2.5 Explainability

Users should be able to see why a district was suggested. An explanation that is convincing but does not agree with the calculation of the score can be misleading. Data quality issues (missing data, approximate travel times, and the incomplete rent coverage) have an impact on reliability, which is why these limits should be explicitly communicated and not be listed as one number.

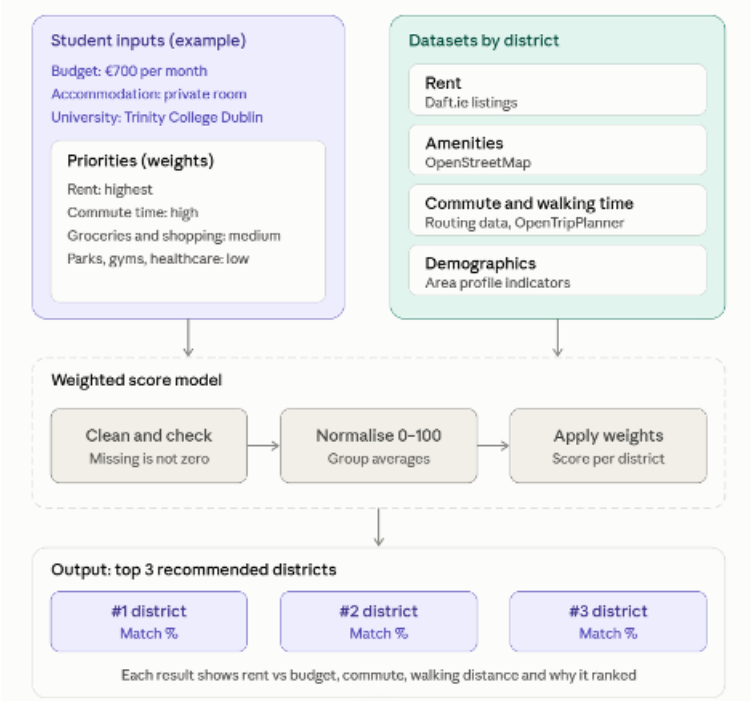



*Figure 5: System Flow*

## 2.3 Existing Systems and their Limitations

There are existing services to explore accommodation, transport, and living costs. But each one deals with a different aspect of the decision-making process. Find Your Dublin aims to bring these factors together to help international students compare residential districts according to their individual needs.

### 2.3.1 Daft.ie

Daft.ie is Ireland's main property website. Users can search for rentals by location, price, number of bedrooms, property type, and can view photos and contact the landlord directly (Daft.ie, n.d.). Very useful when a student is ready to make individual comparisons. It is based on individual properties, however, rather than areas. It does not indicate the quality of the district's accessibility for a student's commute or lifestyle, and students are required to research modes of transportation and surrounding amenities on their own.

Main Features:

- Search for rentals by location, price, and property type.
- See photos and information about the property.
- Write to landlords/letting agents.

Advantage: Helpful for locating and contrasting real properties that are available.

Disadvantage: Absence of a district-level comparison of commute, amenities and personal lifestyle preferences.

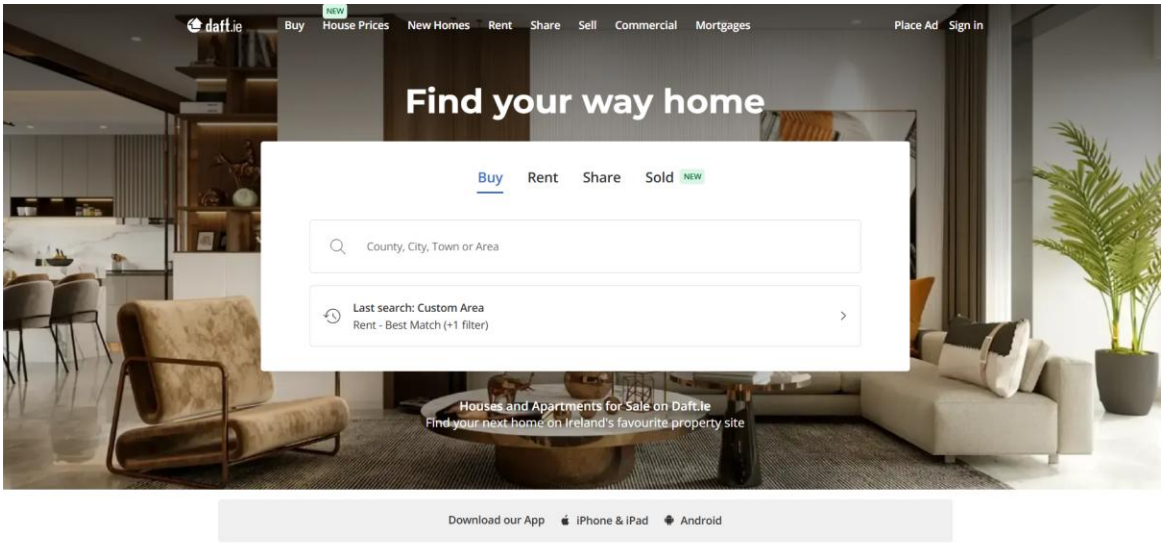



*Figure 6: Daft.ie website*

### 2.3.2 Google maps

Google Maps offers walking, public transport and car directions and estimates the time of a journey, and it displays what businesses and facilities are close to a location (Google, n.d.). This enables students to understand how an area is related to their university. But finding a route does not necessarily equate to determine whether a district is a good place to live. It doesn't provide any data about rent, and students need to manually find all the amenities and add them up.

Main Features:

- Directions for walking, driving, and public transport.
- Estimated journey times and distances.
- Local businesses and facilities.

Advantage: This allows students to evaluate the accessibility between possible residential locations and their university.

Disadvantage: Does not provide a personalized district ranking that combines rental affordability, accommodation preference and amenity data.

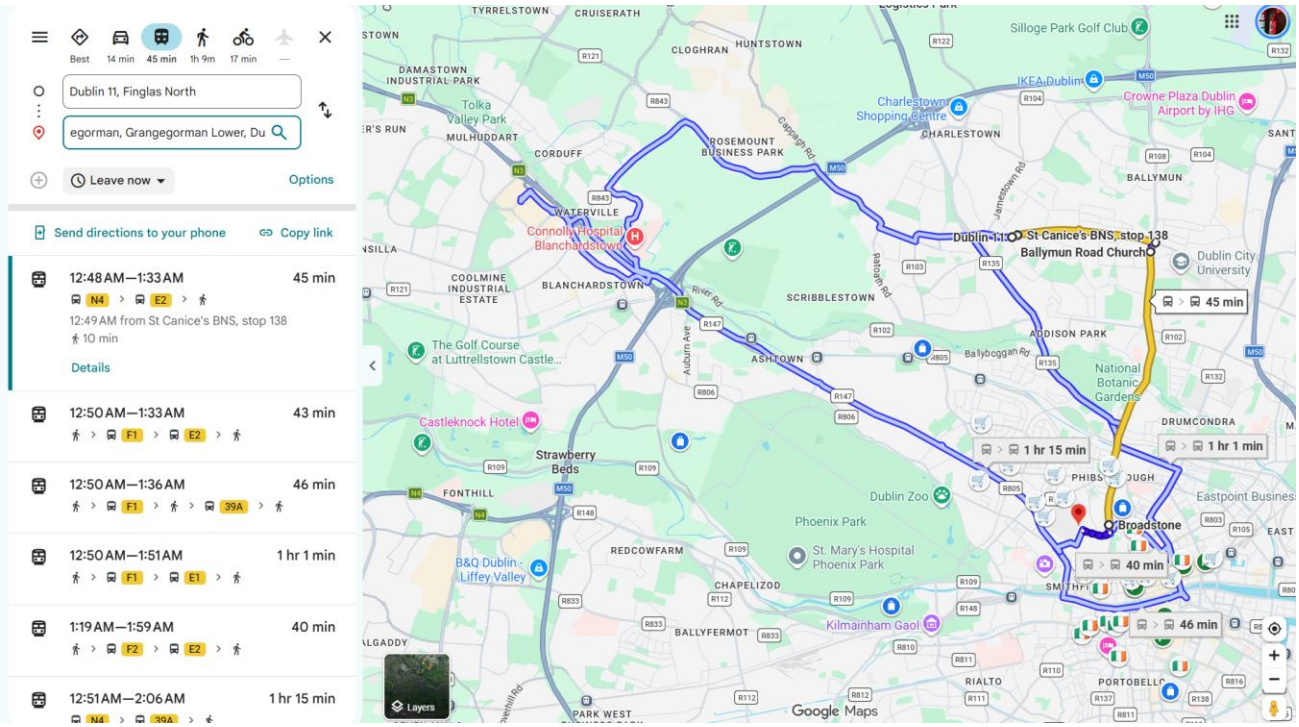



*Figure 7: Google Maps*

### 2.3.3 Numbeo

The cost-of-living data on Numbeo includes average rents, prices of groceries, or transport in cities around the world (Numbeo, n.d.). It is a useful indicator of the average cost of living in Dublin. Most of the data is provided by users and applies to the entire city and does not reflect rent differences between districts or a student's type of accommodation and budget.

Main Features:

- Estimates the cost of living in various cities.
- Rental price and grocery price information.
- Comparisons of everyday expenses.

Advantage: This gives you some useful background information to consider when you're budgeting your move to Dublin.

Disadvantage: City level estimates are not intended to accurately reflect any differences within a district and are not reflective of each individual student's specific type and budget of accommodation.

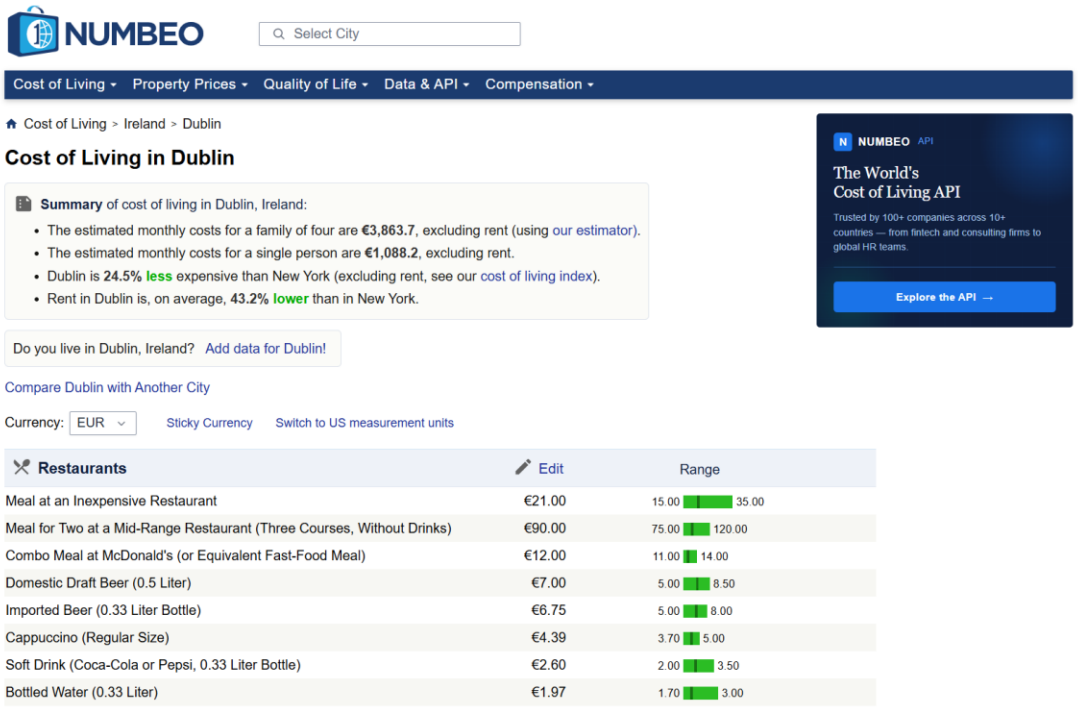



*Figure 8: Numbeo.com website*

Students can search for property on daft.ie, explore routes and facilities around them on Google Maps and find general cost-of-living information on Numbeo. Find Your Dublin is a comparison of residential districts, incorporating affordability, commute, amenity and individual preference, to provide ranked recommendations with explanation. It's an addition to these services, not a replacement for them.

| Feature | Daft.ie | Google Maps | Numbeo | Find your Dublin |
|---|---|---|---|---|
| Purpose | Find properties | Explore routes | Understanding living costs | Compare suitable postal districts |
| Property listings | Yes | No | No | No |
| Local amenities information | Limited | Yes | No | Yes |
| Route and commute information | Limited | Yes | General cost data | Yes |
| Personalization | Listing filters | Manual assessment | General cost data | Top preferences ranking |
| District comparison | Search by location | Map based self-exploration | City level generally | Main purpose |
| Ranked district recommendations | No | No | No | Yes |
| Explanation of recommendation | Property listing details | Manual exploration result | Cost estimates | User preferences-based reasons |

These services are still useful for what they are designed to do. Find Your Dublin's service is filling the gap between rent, commute and amenities – districts can be compared against a student's own needs and there's a clear 'why' for each comparison.

## 2.4 Requirement specification

The requirements for our project were identified based on the needs of international students and the technical objectives of the project. The requirements are prioritised according to their importance and organised into releases across the project timeline.

| Req ID | Name of Req | Description | Priority | Who uses it? | Release/Week | Source |
|---|---|---|---|---|---|---|
| 1 | Calculate distance of commute | Calculates distance from university to reccommending area | A | User | Release 1 (Week 2) | Open Trip Planner, tfigtfs, psm |
| 2 | Collect data for protoytpe from D1 TO D3 | Collect different data like rent, amenities, commute distance, universities | A | Reccommendation Engine | Release 1 (Week 2) | Rent data - RTB, rent.ie, and daft.ie<br><br>Amenitites - OSM (Open Street Map) |
| 3 | Survey TUD students | Collect data from students studying in TUD | A | Crib Crew | Release 1 (Week 2) | Google forms |
| 4 | Reccommending Area | Recommend top 3 dublin areas based on user preference | A | User | Release 1 (Week 3) | With the data collected |
| 5 | Explaining the reccommendation | Explain the shorlist | B | User | Release 1 (Week 3) | Python |
| 6 | Finalise Tech Stack | Choose feasible tools to run the engine | A | Crib Crew | Release 2 (Week 3) | Own research |
| 7 | Collect data for protoytpe from D4 TO D15 | Collect different data like rent, amenities, commute distance, universities | A | Reccommendation Engine | Release 2 (Week 4) | Rent data - RTB, rent.ie, and daft.ie<br><br>Amenitites - OSM (Open Street Map) |

| | | | | | | |
|---|---|---|---|---|---|---|
| 8 | Create UI design | Basic UI Building | A | User | Release 3 (Week 5) | Node.js |
| 9 | Map view for district information | Create map view for different dublin districts | B | User | Release 3 (Week 5) | Node.js |
| 10 | Improve code with new data | Update datasets in py file, and improve code based on it. | A | User | Release 3 (Week 5) | Python |
| 11 | Build recommendation api | convert the recommendation engine into web api that website can send user preferences and receive the top 3 areas | A | Reccommendation Engine | Release 3 (Week 5) | Python |
| 12 | Host Backend on Azure | Deployed the api on azure app so its available with public utl | A | User | Release 3 (Week 5) | azure |
| 13 | Automatic deployment | automatically redeploy the backend when the code is pushed to the main branch of github | B | Crib Crew | Release 3 (Week 5) | github |
| 14 | Connect frontend to backend | website sends users priority to api and displays the result | A | User | Release 3 (Week 6) | Node.js |
| 15 | Validate user input | Reject invalid input and display error message | B | User | Release 3 (Week 6) | Python |
| 16 | Handle missing area data | give a heads up to the user when an area is left out because data for their chosen accomodation type is missing | B | User | Release 3 (Week 6) | Python |
| 17 | Find supermarkets in a particular area | Investigate and confirm number of supermarkets in each area | B | User | Release 3 (Week 6) | Manual verifiaction |
| 18 | Find parks, gyms | Investigate and confirm number of parks in each area | B | User | Release 3 (Week 6) | Manual verifiaction |

| | | | | | | |
|---|---|---|---|---|---|---|
| 19 | Find cafes/restaurants | Investigate and confirm different types of cafes/restaurants in each area | B | User | Release 3 (Week 6) | Manual verifiaction |
| 20 | Find pharmacies and hospitals | Investigate and confirm number of pharmacies and hospitals in each area | B | User | Release 3 (Week 6) | Manual verifiaction |
| 21 | Display area demographics | Display useful Census demographic information about each area | B | User | Release 4 (Week 7) | Census dataset (CSO) |
| 22 | International community | Show demographic community information from Census data | B | User | Release 4 (Week 7) | CSO |
| 23 | Mobile responsive UI | Website properly works on phone,tablets,desktop | A | User | Release 4 (Week 8) | Non functional |
| 24 | Make website more attractive | Design website attractive and attention grabbing | A | User | Release 4 (Week 8) | Non functional |
| 25 | Initiate AI agent | Introduce AI chatbot for immediate info | C | User | Release 5 (Week 9) | |
| 26 | Add Voice Assistant function | Introduce Voice Assistant | C | User | Release 5 (Week 9) | |

# 3. Technical Solutions:

## 3.1 What does the system do?

Our system is a web application that helps students, particularly international students new to Dublin to find which part of the city would suit them before they start searching for an accommodation. Rather than recommending an individual house or apartment, it compares Dublin's postal districts and suggests three best matches based on inputs. After logging in, the student enters their university, monthly budget and type of accommodation they prefer (shared room, private room, or one or two bedroom apartment and student accommodation), and then rates how important are these six factors to them: rent, commute time, shopping and groceries, restaurants, healthcare, and recreation and wellness. For each recommended area the system shows a match score, the average rent, the public transport commute and walking distance to their campus from respective locations, and a short explanation of why the area ranked where it did, since explanations make recommendations easier to trust (Zhang and Chen, 2020) followed by brief profile of the area based on census data and a link to a district profile page with an interactive map.

The area suggestions come from a weighted, and rank based scoring model, a common approach in multi-criteria decision analysis (Malczewski, 2006). Each rating is converted into weights: 'very important' counts as 3, 'important' as 2, 'slightly important' as 1 and 'not important' as 0, so a factor students doesn't really care about has no effect. The areas are ranked on each factor and given a standardised score from 0 to 1, where best match scores 1 and worst scores 0. Lower score is better for rent and commute, higher is better for amenities.

We chose to use ranks rather than raw values, to prevent the outcome being skewed by the performance of one well-served district. Amenities are currently rated in four groups such as groceries, restaurants, healthcare and recreation and wellness. Within each group amenities can be ranked separately and the group

score is their average, so common amenities like pharmacies cannot outweight hospitals, we extend the form so that student can rate amenities separately, for example a hospital but not pharmacies. The weighted scores are added together and shown as percentage to give match score. Budget acts as a warning rather than filter, if none of the top three areas are affordable, the system shows a message and points to the best one. This idea is similar to the housing recommender by Daly et al. (2014), which weighs rent against journey times to places the user chooses; we apply it at the level of postal districts

## 3.2 System Architecture

Fig 6 shows the overall architecture of the system. Student's browser loads the front end which gets the input from student and sends it to Python API. The API passes the inputs to the recommendation engine and runs against the data we have provided and returns the results to the front end to display (Malczewski, 2006). Data are cleaned and combined in a separate step before it reaches database, so the live application only reads the prepared data apart from storing user accounts for login. Everything is hosted on Microsoft Azure, previously we planned to use Netlify, Neon and Render, but our mentors pointed out these free services were slow, so we researched about the alternatives and moved to Microsoft Azure using the free

credit available on Azure for students.

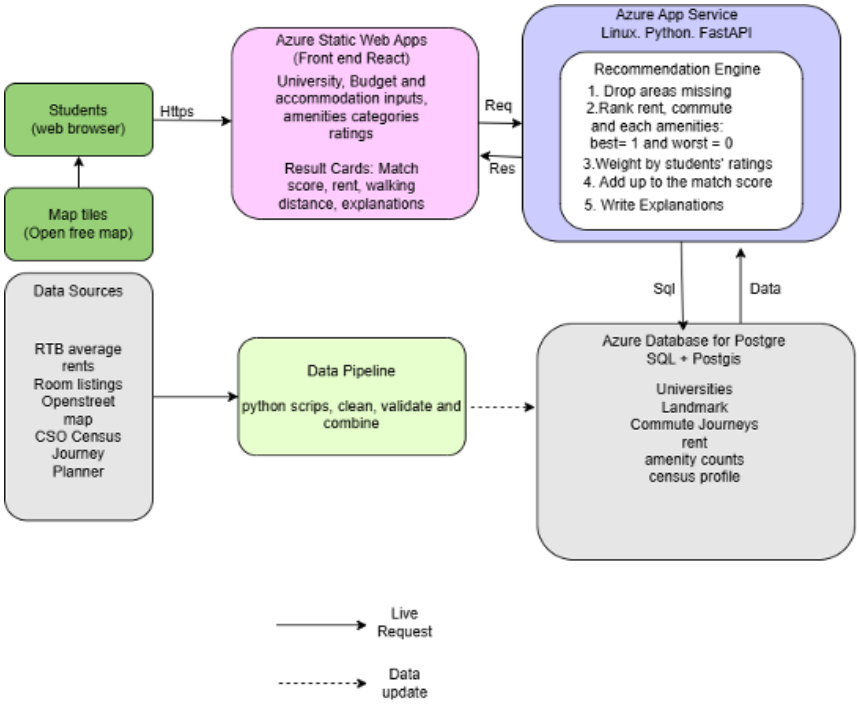



*Figure 9: System Architecture*

## 3.3 Front end

The front end takes the students step by step throughout the search. After logging in, they enter their accommodation budget, name of the university from the list of college across Dublin, and select the accommodation type they want.

The front end is built with React and Vite. We chose React because the site is made of repeated parts, such as the rating cards and the results cards, which can be written once as components and reused, and React Router lets students move between the login, form, results and district profile pages without the page reloading. Vite was used because it starts and rebuilds the site quickly during development and produces small files for hosting. The interactive map on the district profile page uses MapLibre with OpenFreeMap map tiles, which are free and based on OpenStreetMap, so no paid map service is needed.

On the next page, they rate how important the rent, commute time and each group of amenities are, using the four options from ‘very important’ to ‘not important’. The results page then shows up the top 3 matches, with its match score, average rent, walking distance from the respective location to university, how far it is under or over the budget, commute time to the campus, an explanation of why this area is best suited (Zhang and Chen, 2020) and a short profile of the area. From each recommendation, the student can view a district profile page with an interactive map of Dublin’s postal districts, which shows selected district’s rent, amenities information and famous places of the postal district. A login was added so that students can return to their results later.

## 3.4 Back end

We use Python FastAPI to write the back end with pandas and NumPy to do the scoring. It has minimal number of endpoints, one return the options for the form, one gets input from the student and returns the recommendations, and confirms the data has loaded. We chose FastAPI because our recommendation logic was already written in Python with pandas, so the scoring and the API share one language. FastAPI is also lightweight and fast, and it creates API documentation automatically, which we have found useful for testing. The API runs on Azure App service on Linux. We use the free F1 tier during development and the basic B1 tier for demonstrations, because B1 keeps the application running continuously and avoids the delay on first response. For storage we are using Azure Database for PostgreSQL with the PostGIS extension. We chose PostgreSQL because our data is structured tables linking districts, universities and journeys, and PostGIS adds location support, which supports location based queries such as finding which points lie close to the campus. We have built and tested a script that loads our data into tables for universities, commute times, along with summary view of median commute for each district and the university.

## 3.5 Data collection and Preparation

Data were collected from multiple sources to develop a comprehensive dataset for Dublin postal districts from D1 to D24. The collected data cover attributes such as rental prices, accommodation options, local amenities, commuting information, and characteristics of the area. The datasets were cleaned, processed, and transferred into Excel and CSV files for integration and use in the recommendation engine.

**Rent Data:**

Rental data for one, two, and three bedroom apartments were collected from the Residential Tenancies Board (RTB), which provides average rent by area, property type, number of bedrooms, and quarter/year. During preprocessing, the dataset was cleaned and missing values were handled. The missing rental values were estimated by calculating the mean of available rental values from nearby areas with comparable property types and bedroom categories. This helped to fix the gaps and consistency across districts. Additionally, data on single-sharing and double –sharing accommodation were collected manually from Daft.ie and Rent.ie. The collected data were transferred into Excel files according to district and accommodation type.

**Amenities Data:**

Amenity data were collected from OpenStreetMap using Overpass API. Queries were performed for each postal district from D1 to D24 to identify local amenities such as supermarkets, grocery shops, restaurants, cafes, gyms, pharmacies, hospitals, parks, and shopping malls. The retrieved data were processed and grouped by district to determine the availability and distribution of local amenities.

**Commute Data:**

Five representative landmarks coordinates were selected within each Dublin Postal district and validated aganist the Dublin Postal district GeoJSON boundaries to ensure they fell within respective districts. For

example, D1 landmarks includes landmark such as Custom house Quay, O'connell Street, Jervis Street, and Mountjoy Square, along with another representative point. OpenStreetMap pedestrian-network data and TFIGTFS data were loaded into OpenTripPlanner to build a routing graph. Python script automated the process by passing the landmark coordinates of TU Dublin Grangegorman to the OTP routing server. OTP calculated the routes and returned journey distances and travel times. The results collected were organized by districts for comparision and use in recommendation engine.

**Census Data:**

Demographic data were collected from CSO Census 2022 dataset. Relevant information included population distribution, age groups, and nationality characteristics at the available geographic level. When necessary, electoral division data were mapped to the corresponding Dublin postal districts to support district-level comparisons. The collected imformation was cleaned and transferred into Excel files for demographic analysis.

**Data Organization and Preparation:**

Finally, the collected datasets were organized into structured Excel and CSV files covering rent, amenities, commute, and census information. The data were prepared for integrating into the recommendation engine, enabling comparision between accommodation budget, commute time and distance, local amenities, and demographic characteristics across Dublin postal districts from D1 – D24.

supermarkets, grocery shops, shopping malls, restaurants, hospitals, pharmacies, gyms and parks was collected from OpenStreetMap and counted for each district (Haklay, 2010), now it covers all the amenities data for 21 Dublin postal districts. Population, age, housing type come from the CSO 2022 Census dataset. For commuting, we chose about five landmarks in each district and recorded the public transport journeys from each to fifteen universities and colleges, all leaving at 8 am on weekday, contains 1,635 journeys (Daly et al., 2014). Walking distance and times to each campus from the respective locations were collected in the same way. We also conducted a survey among TU Dublin students about where they live, which amenities they use and how they travel, and will use the results to check our factors reflect what students value the most (Jaroszewicz and Majewska, 2021). The data is currently held in CSV and Excel files and is being moved into PostgreSQL. We considered adding a crime score, but Irish crime figures are published by Garda station rather than postal district, so this has been left for future work.

## 3.6 Prototype

Figure 7 shows the home page of our demo. It welcomes international students and explains in one line what the site does. Helping students find a Dublin district that fits their budget, commute and everyday needs. The navigation bar gives access to the main parts of the site such as exploring districts, how the recommendation words, logging in, and the "Find my match" button, which takes the student to the preference form. This site plays a short video of Dublin, showing famous places across Dublin, making users more comfortable. This site also shows a list of universities across Dublin where all the universities are linked to their respective sites.

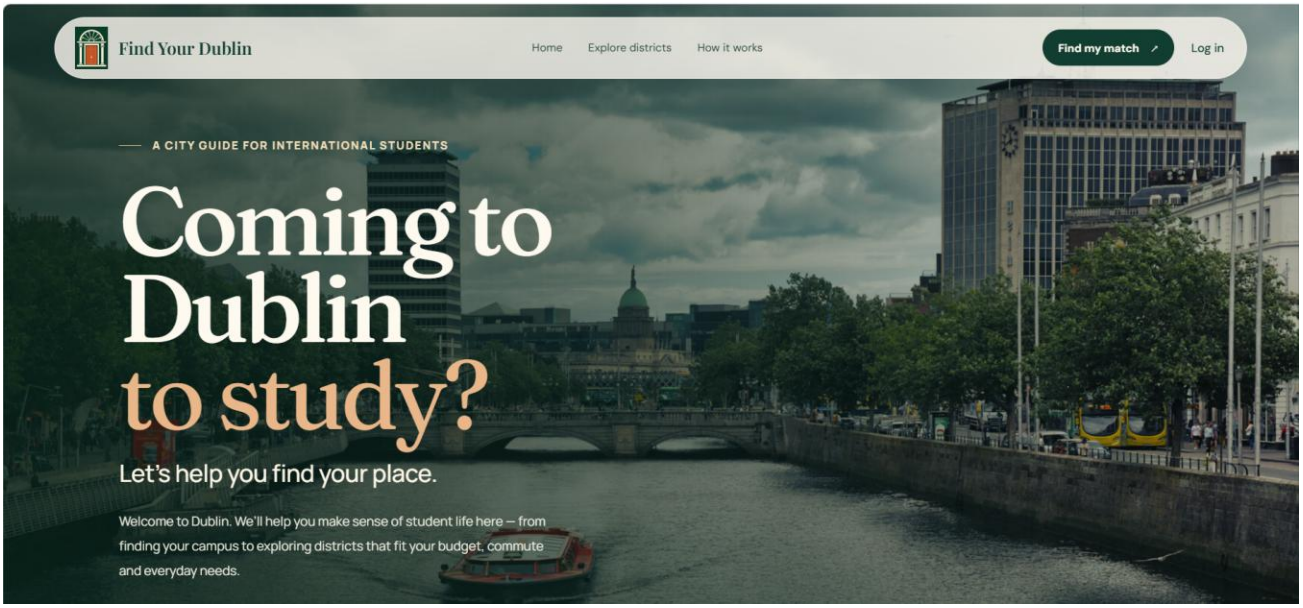



*Figure 10: Home page*

Figure 8 shows the first step of the preference form. A progress bar at the top tells the student they are on step 1 of 3. They set their monthly rent budget using a slider from 400 euros to more than 2500 euros, further down the same page, they chose their university and type of accommodation they prefer, before moving on to rate their priorities.

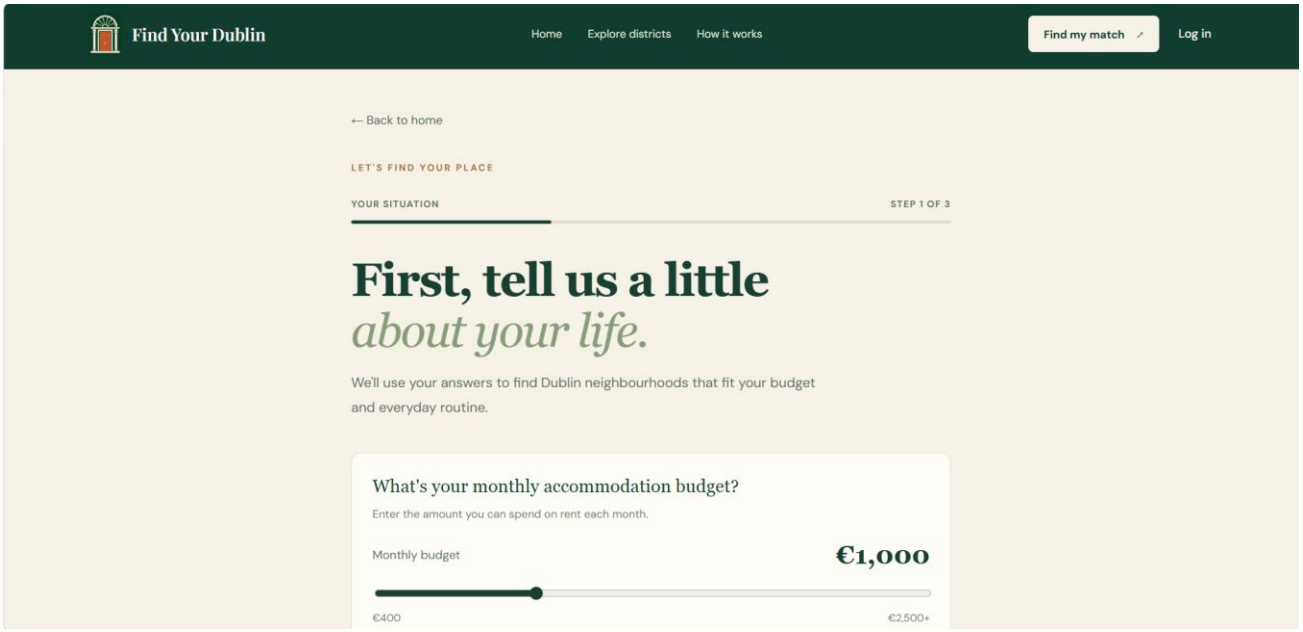



*Figure 11: Entering student's monthly budget*

Figure 8 shows the second step, where students can set priorities to the factors. Each factor (rent, commute, shopping and groceries, restaurants, healthcare, and recreation and wellness) has its own card with four options, from 'Very Important' to 'Not Important', and 'Important' selected by default. These choices become the weights used to rank the district, 'Very Important' counts the most and 'Not Important' removes the factor from the score. A "Back to your situation" link the lets the student return to step 1 and change their answers.

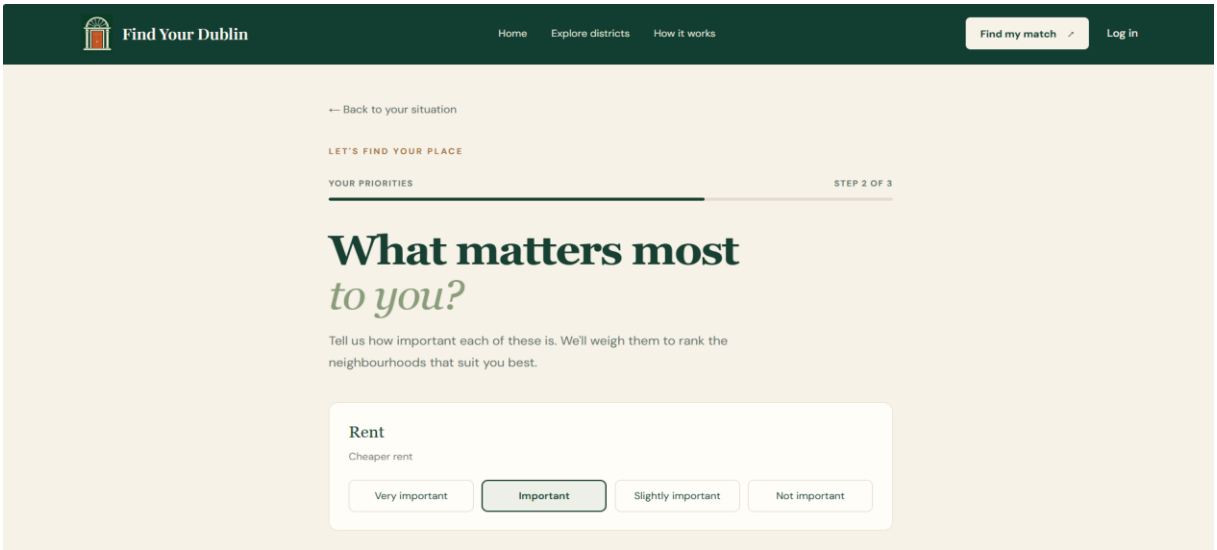



*Figure 12: Rating importance of each factor*

Figure 9 shows the results page for a student at Dublin Buisness School looking for a one-bedroom apartment with a budget of 1700 euros per month. The summary at the top confirms their answer and show that 19 areas were compared. Each recommended district appears as a card. Here D1 (North Inner City) is ranked first with 86% match score. The card shows the average rent, the public transport commute and the average walking distance to the campus, which can be expanded to show the walking distance to Dublin Business School from each place in the district. "Why D1 is #1" section explains the ranking in plain language, for example its short commute and the number of supermarkets. Every card comes with "View District Profile" button where students can view the map of the district.

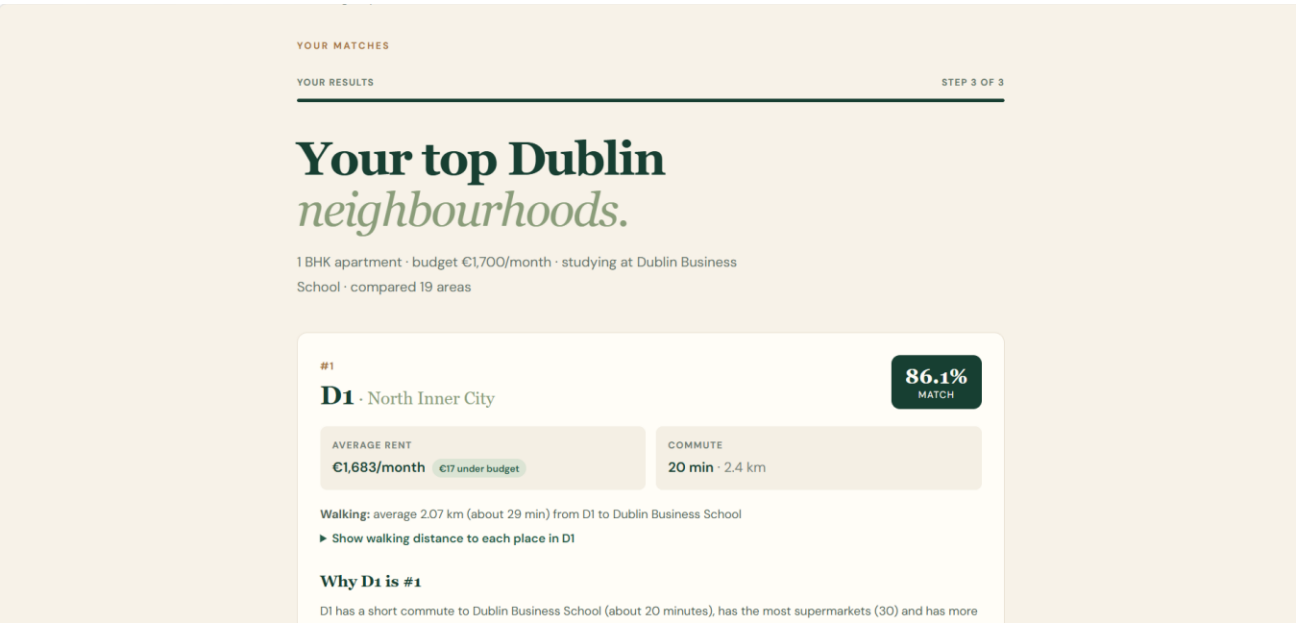



*Figure 13: Result page showing top 3 recommendations*

Figure 10 shows the district profile page, which students reach from their results page. On the left, an interactive map of Dublin's postal districts lets them zoom, move around and select a district, with the chosen district highlighted. On the right, the profile shows the selected district's average rent for one and two bedroom apartments and how many of each amenity it has, such as groceries, restaurants, cafes, gyms, parks and pharmacies and list famous places of that respective district. This lets students explore an area in more detail before deciding whether to look for accommodation there.

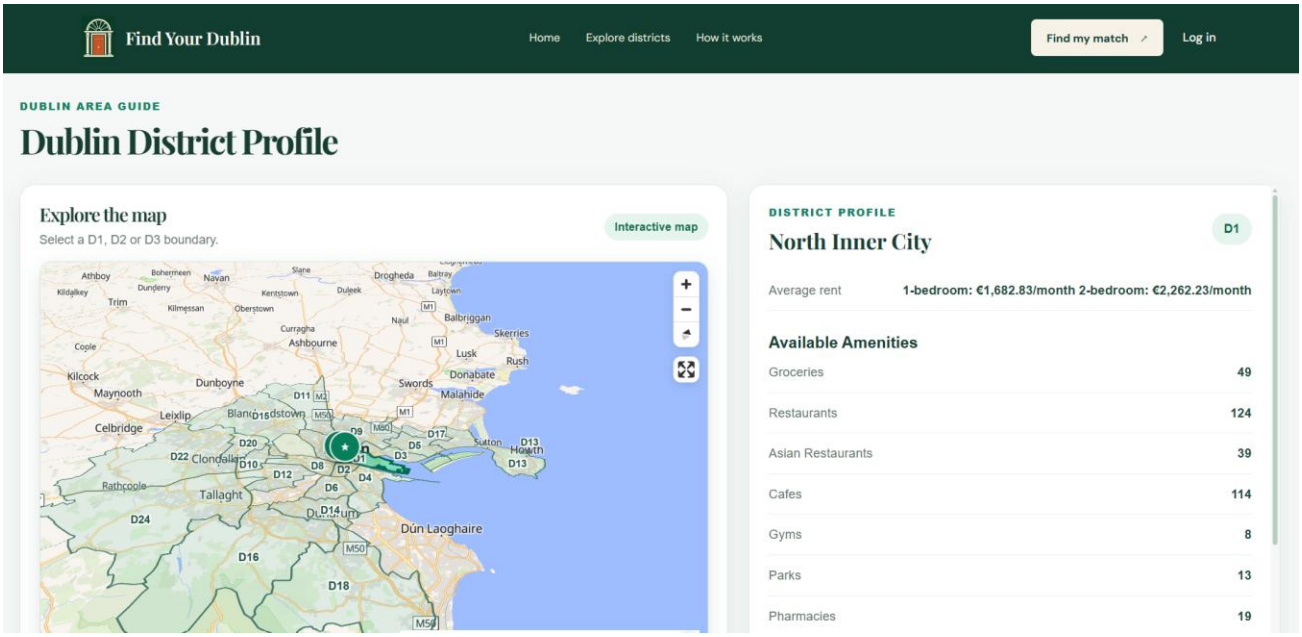



*Figure 14: District profile page*

# 4. Evaluation

## 4.1 What does success look like for your system?

Students in Dublin gathers information about an area from daft.ie, rent.ie, reddit forums, Facebook groups and YouTube videos. But the information provided is based on the particular persons preferences and takes time to put together by pulling it off from the platform. Our system succeeds if it provides the user same information much faster and in much reliable way. We define success by measuring the following goals.

| Goals | Measures | Target |
|---|---|---|
| Saving users time | Time taken to reach a decision | User should able to make decision at least 30% fast |
| Maintaining the users budget boundaries | Top 3 recommendations should be close to users budget | 100% of the test cases |
| Accuracy in response | All the displayed data to be verified by official sources | Commute time should be approx 5 mins up or down. Rent should be within 10% of buffer margin |
| Correct Ranking | Test cases with already known answers | All test cases should be cleared |
| Users agreement with results | Participants should agree with the top recommendations | At least 75 to 80% |
| Easy to use | System usability scale score | Should be above 68 (Brooke, 1996; Sauro and Lewis, 2016) |
| Responsive | Api response time on azure | Around 2 seconds |

## 4.2 How will you evaluate the system that you built?

We will evaluate the system by dissecting it in several parts. We took a survey of 59 students across TUD campus asking them about their area like where do they live? How do they commute to university? What amenities do they have nearby? The survey gave us important insights like 56% of total students selected Bus 53% selected Luas and 30% train which is why we decided to use TFI for calculating commute time.

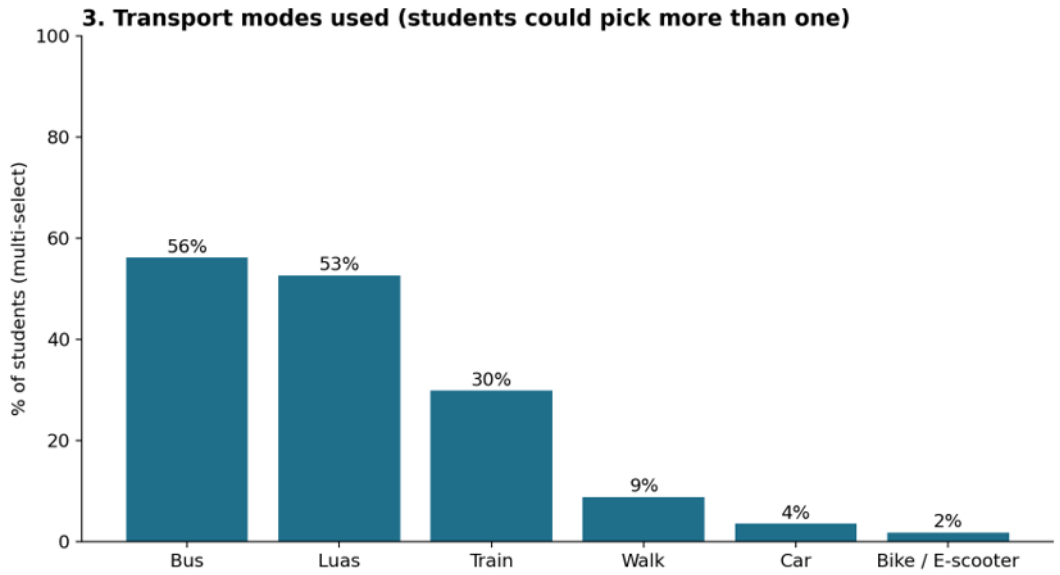



*Figure 15: student's preferred mode of transport to TUD*

In this survey we also found out that more than half of the students have a commute time of 45 mins or longer to reach TUD grangegorman. The median commute time is 50 minutes and about one in three travels for around 1 hour. Area choice does affect the lifestyle of students. Longer commute time takes away the time form study, work and social life. It also add to the expenses of travel cost. Hence including commute time in our recommendation system and evaluating it would be key.

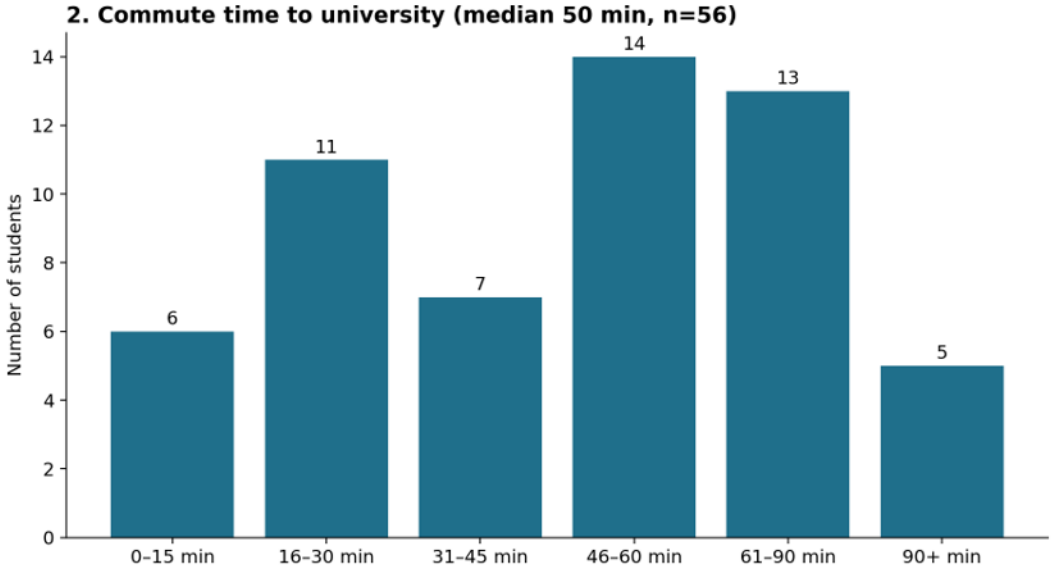



*Figure 16: student's commute time to TUD*

So for data evaluation we will check commute time on TFI website and google maps. We will refer it for sample routes between each landmark and universities. We also corrected the landmarks which were found in wrong district during development process. We will compare our daft.ie rent average with RTB figures if there are very few listings in particular area. Amenity count will be checked in person or street view via google map.

We will write test cases with known answers to evaluate algorithm. If users priority is rent then area with cheapest rent should come out first. If no area is within users budget the system should warn the user. If commute is priority area closest to the university should show up. Explanation for ranking the particular area should make sense with respect to the score.

We will performance test by sending 50 to 60 requests to the Api hosted on azure and record the fastest, average and slowest response time. It should be within our target that is around 2 seconds. The cold start will be measured separately.

We will let the actual students test our whole system. Mainly the students who are new to Dublin will be ideal candidate for this test. We will provide students with scenario or use their own budget and location that they think is ideal their accommodation. At first we will let them handle the task as they wish they can use their own sources. Later we will let them use our system. We will record the time taken to reach the conclusion by each method. Students who participated will be given system usability scale questions (Bangor, Kortum, and Miller, 2009) (ISO, 2018). Feedback will be collected from the student whether they agree with the top recommendation by our system. How confident they feel after using our system? What changes would they like to make in our system or what feature they want to add?

After the user test we will analyze the result by comparing the average decision time taken by the user without using our system and while using our system. We will also take the percent of students who agree with our systems top recommendation. We will group feedback like common issues, missing factors or unclear explanation. It will all be fixed in final demo.

# 5. Conclusion: The Plan

Our project management strategy follows the agile methodology where we develop the project in small stages every week, test the work regularly, get feedback and make the necessary improvements throughout the process. We use Jira to assign tasks, track progress and monitor deadlines. Our team lead rotates every week allowing other members to take responsibility of coordinating the tasks. Every week Friday, we present our work to our mentors, receive their suggestions and feedbacks and discuss the next steps. After the presentation on Friday we meet face to face to discuss our mentor's feedback, identify the challenges, plan the next steps to refine the project and progress it. This iterative approach allows us to adapt to challenges while maintaining steady progress towards our goals.

So far, we have made progress in collecting and preprocessing the rental data from RTB, rent.ie, daft.ie and amenities data. We automated the commute distance calculation. Then we built the recommendation system to consider user's preferences. We have built the frontend and backend. Next we have to integrate both and deploy it into a complete website.

One of the main challenges we faced was data collection. The RTB rental datasets contained many missing values, some districts had limited data making it difficult to maintain consistency. Therefore, preprocessing, cleaning and validating the dataset was required. Another major challenge was calculating the commute distance and walking distance from various points in a district to multiple universities. As this would be a time consuming task to do manually, we automated it using Python and OpenTripPlanner which helped us to calculate distance and time for all the universities in Dublin and organize it into a csv file to load it to our recommendation engine.

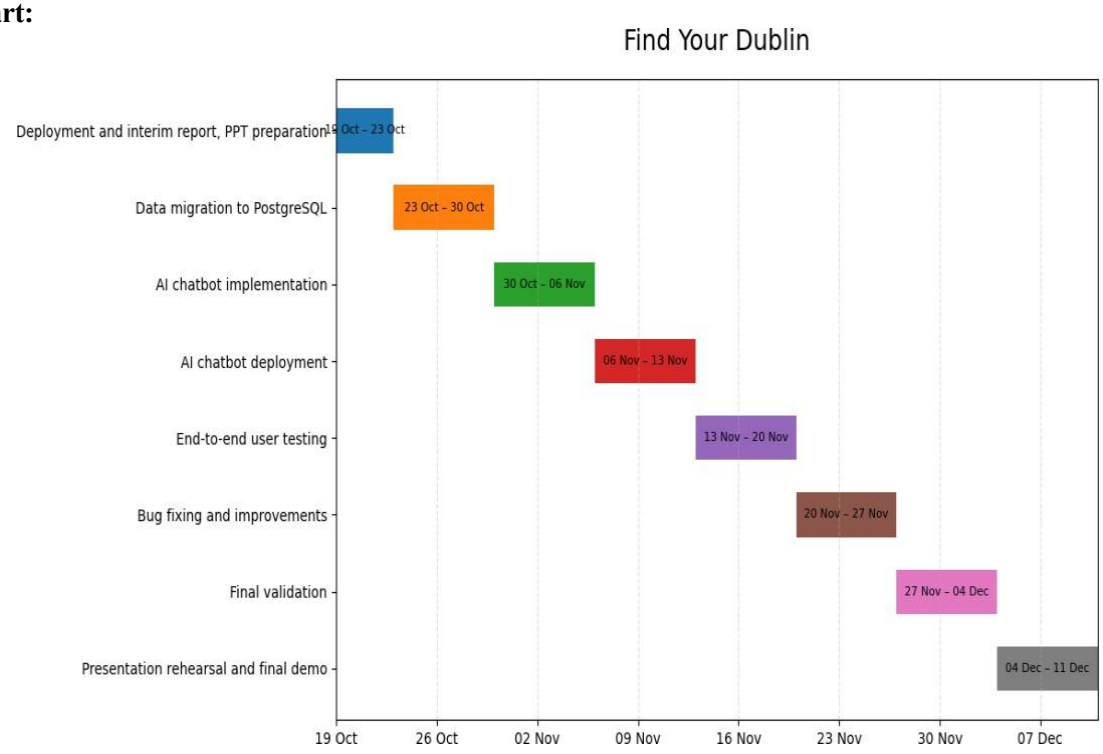
Developing an effective recommendation algorithm was another main challenge. Students have different priorities including amenities nearby, rent and commuting distance to the university. So our recommendation engine must use these factors to rank the areas correctly according to the user's preferences. Integrating the datasets, recommendation engine, backend, and frontend into a single website also requires coordination among team members. Another challenge is deploying the website on a reliable hosting platform that can handle user requests efficiently and provide faster responses, ensuring a smooth user experience.

During the remaining time, we will continue following our Agile approach by setting weekly goals, presenting our progress and incorporating mentor's feedback alongside our own ideas. One planned enhancement is to implement an AI chatbot in our website to help users get information about Dublin districts. For example, users could ask, "Which areas have monthly rents below 750 euros, are close to the TU Dublin Grangegorman campus and have gyms nearby?", "How many parks are there in Dublin 1?" or "Are there any Tesco supermarkets in Dublin 4?" The chatbot is intended to make the recommendation system more interactive. This will be implemented on the upcoming weeks.

Our immediate priorities are to integrate the frontend and backend and deploy it on Azure and minimise the response times. We will then focus on testing the system with different user preferences, assessing the recommendations and verifying the budget constraints are strictly enforced. Following this we will conduct end to end testing, debug any issues and do the documentation. We aim to deliver a reliable, user friendly recommendation system that helps international students to find their suitable area based on their budget,

preferences when they come to Dublin.

**Gantt chart:**



# References


1. Connor, D. (2025) ‘International student's search for accommodation an “emotional rollercoaster”’, RTÉ News, 5 September. https://www.rte.ie/news/ireland/2025/0905/1531901-student-accommodation-dublin/
2. Higher Education Authority (2025) Key Facts & Figures: 2024/25. https://hea.ie/statistics/data-for-download-and-visualisations/key-facts-figures-report/
3. Daft.ie (n.d.) Property for rent. https://www.daft.ie/
4. Google (n.d.) Google Maps. https://maps.google.com/
5. Numbeo (n.d.) Cost of Living. https://www.numbeo.com/cost-of-living/
6. OpenStreetMap contributors (n.d.) OpenStreetMap. https://www.openstreetmap.org/
7. **Daly et al. (2014)**, Multi-criteria journey aware housing recommender system: https://doi.org/10.1145/2645710.2645764
8. **Jaroszewicz & Majewska (2021)**, Group spatial preferences of residential locations (free PDF): https://www.mdpi.com/2071-1050/13/9/4690
9. **Malczewski (2006)**, GIS-based multicriteria decision analysis: https://doi.org/10.1080/13658810600661508
10. **Zhang & Chen (2020)**, Explainable recommendation: publisher page: https://www.nowpublishers.com/article/Details/INR-066 free PDF on arXiv: https://arxiv.org/pdf/1804.11192
11. **Haklay (2010)**, How good is volunteered geographical information?: https://journals.sagepub.com/doi/10.1068/b35097
12. Brooke, J. (1996) ‘SUS: A “quick and dirty” usability scale’, in Jordan, P.W., Thomas, B., Weerdmeester, B.A. and McClelland, I.L. (eds.) Usability Evaluation in Industry. London: Taylor & Francis, pp. 189–194.
13. Bangor, A., Kortum, P. and Miller, J. (2009) ‘Determining what individual SUS scores mean: adding an adjective rating scale’, Journal of Usability Studies, 4(3), pp. 114–123.
14. Sauro, J. and Lewis, J.R. (2016) Quantifying the User Experience: Practical Statistics for User Research. 2nd edn. Cambridge, MA: Morgan Kaufmann.
15. ISO (2018) ISO 9241-11:2018 Ergonomics of human-system interaction – Part 11: Usability: Definitions and concepts. Geneva: International Organization for Standardization.
16. Irish Council for International Students (ICOS) (2025) Renting in Ireland: The Housing Crisis from the Perspective of International Students. Available at: (Accessed: 21 October 2026).
17. Residential Tenancies Board (RTB) (n.d.) RTB/ESRI Rent Index - Data Set. Available at: (Accessed: 21 October 2026).
18. Technological University Dublin (n.d.) Accommodation and Living in Dublin. Available at: (Accessed: 21 October 2026).